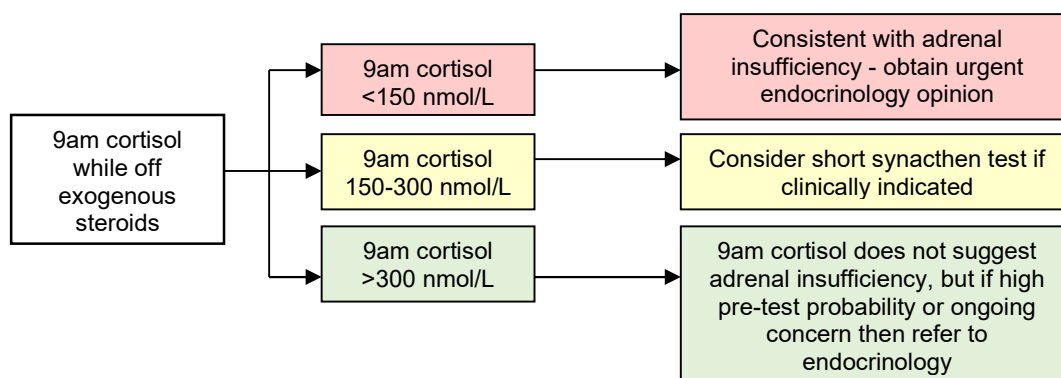


Derbyshire Shared Care Pathology Guidelines

Short Synacthen Test (in adults) for Primary Care

The short Synacthen test is a simple procedure for investigating reduced adreno-cortical function and adrenocortical reserve. It is a preliminary test and abnormal responses need to be followed up with further tests and an endocrinology opinion should be sought.



Synacthen (Tetracosactrin) is a synthetic analogue, comprising amino acids 1-24 of the 39 amino acid peptide Adenocorticotrophic Hormone (ACTH). This sequence retains the full biological activity of intact ACTH. Synacthen stimulates the normal adrenal cortex to secrete cortisol, which can then be measured in serum.

It should be noted that Prednisolone and hydrocortisone cross react with cortisol assays (i.e. can give falsely raised cortisol levels), but dexamethasone and budesonide do not. The short synacthen test is only suitable for patients that have been off exogenous steroids for an appropriate length of time – see the section on 'patient preparation' for more information.

NICE guideline [NG243 Published: 28 August 2024] recommend that patients on long term steroids do not usually require a Synacthen test, but rather a glucocorticoid steroid dose-tapering regimen.

The oral contraceptive pill and other oral (not transdermal) oestrogen replacement therapies can cause cortisol levels to falsely be increased by up to 50%. Therefore caution is required when interpreting results on oral oestrogens. Where possible oral oestrogens should be replaced by transdermal oestrogens for 6 weeks before testing. Where this is not practical then, clinician should consult Endocrinology.

CONTRAINDICATIONS

- Pregnancy
- Previous hypersensitivity to ACTH, Synacthen or Synacthen depot
- The first 6 weeks following pituitary surgery

SIDE EFFECTS

Local or systemic hypersensitivity reactions have been reported very rarely following Synacthen injection, particularly in children with a history of allergic disorders.

PRECAUTIONS

Patient should be kept under observation throughout the period of this test.

PREPARATION

Planning

The test can be carried out at any time, but **should ideally be performed as near to 9am as possible**. Cortisol levels decline throughout the day and cortisol responses between the morning and late afternoon may differ by as much as 100 nmol/L at 30 minutes post Synacthen, making interpretation of afternoon tests difficult if the response in cortisol is abnormal. An adequate response to synacthen is a valid result at any time of day.

Patient

- Fasting is not necessary
- Patients on exogenous steroids should withhold their steroid for the following times, but take them immediately after completing the synacthen test:
 - Hydrocortisone - withhold for 18hrs before the test
 - Prednisolone - withhold for 24hrs before the test
 - Topical steroids and inhaled steroids - withhold for 48hr prior to the test
 - Steroid joint injections - withhold for 8 weeks prior to the test
- Patients on dexamethasone or budesonide should be discussed with the endocrine team before testing
- All medication should be noted on the request form

Equipment

Obtain the Synacthen from a Pharmacy or Chemist (1 mL ampule, containing 250 micrograms). You will also require 2 SST (yellow top) vacutainer tubes.

PROCEDURE

Samples must be clearly labelled with patient name, date and time, eg 09:30

Time (mins)	Test	Tube type
Basal sample	Cortisol	SST (yellow top)
0 min	Inject 250 micrograms of Synacthen i.m. or i.v.	
30 min	Cortisol	SST (yellow top)

Send both cortisol samples together with a completed Chemical Pathology request form to the Chemical Pathology Department as soon as the test is finished.

INTERPRETATION

A **NORMAL** response is defined as a 30 minute serum cortisol concentration greater than 420 nmol/L.

Consideration of the increment in cortisol can add to interpretation in some circumstances but a high baseline cortisol due to concurrent stress may often preclude a further significant rise in cortisol without indicating adrenal insufficiency.

A normal result excludes primary adrenocortical insufficiency, but **does not necessarily exclude ACTH deficiency**. Partial ACTH deficiency may result in a normal or reduced response to Synacthen.

A **DECREASED** response (30 minute cortisol < 420 nmol/L) may indicate:

1. Primary adrenal failure (such as Addison's disease). Results typically show a low baseline cortisol with little or no response to Synacthen
2. Adrenal atrophy secondary to prolonged ACTH deficiency
3. Adrenal atrophy secondary to long term steroid therapy (including topical, nasal or inhaled steroids) (caution if interpreting result in a patient who has recently been on supraphysiological doses of glucocorticoid)

Abnormal responses need to be followed up with further tests and an endocrinology opinion should be sought through A&G. An urgent referral by contacting the Endocrinology on-call team is required if the 0min result is <100nmol/L or the 30min result is <250nmol/L and the patient is not already on exogenous steroids.

Authors: Biochemistry Standard Clinical Guideline Group, October 1995		
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Dr R Stanworth, Mrs H Seddon	Nov 2015	30 th Nov 2018
Biochemistry/Endocrinology MDT meeting	23/01/19	30/04/22
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Dr A Ugur, Dr D Hughes, Dr A Pillai, Sarah Knowles, Dr P Blackwell	Jan 2025	31/01/2028